Problem 25.2

Find the force of interest at which
$$\bar{s}_{\overline{20}|} = 3\bar{s}_{\overline{10}|}.$$

Solution

For a continuous annuity accumulated for n years at a constant force of interest δ,
$$\bar{s}_{\overline{n}|} = \frac{e^{\delta n} - 1}{\delta}.$$
Let
$$1 + i = e^{\delta}.$$
Then
$$e^{20\delta} = (1+i)^{20} \qquad \text{and} \qquad e^{10\delta} = (1+i)^{10}.$$

The given equation is
$$\bar{s}_{\overline{20}|} = 3\bar{s}_{\overline{10}|}.$$
Substitute the formula for $\bar{s}_{\overline{n}|}$:
$$\frac{(1+i)^{20} - 1}{\delta} = 3\left(\frac{(1+i)^{10} - 1}{\delta}\right).$$
Cancel $1/\delta$:
$$(1+i)^{20} - 1 = 3\big((1+i)^{10} - 1\big).$$
Let
$$x = (1+i)^{10}.$$
Then the equation becomes
$$x^2 - 1 = 3(x-1).$$
Simplifying,
$$x^2 - 3x + 2 = 0.$$
Factor:
$$(x-1)(x-2) = 0.$$
Since a positive force of interest is intended, use
$$x = 2.$$
Therefore,
$$(1+i)^{10} = 2.$$
But $1 + i = e^{\delta}$, so
$$e^{10\delta} = 2.$$

Taking natural logarithms,

$$10\delta = \ln 2.$$

Thus,

$$\delta = \frac{\ln 2}{10} = 0.0693147.$$

Therefore, the force of interest is

$$\boxed{6.93147\%}.$$